

Session Documentation – Git & GitHub Workflow

Session Details

- **Session Title:** Git & GitHub Basics
 - **Date:** 6 February 2026
-

Objective of the Session

The objective of this session was to help students understand:

- Git installation and basic configuration
 - Setting up SSH keys for secure GitHub access
 - Pushing local projects to GitHub
 - Using Pull Requests for team collaboration
-

Topics Covered

- Git installation
 - Git configuration
 - SSH key generation
 - Connecting Git with GitHub
 - Creating a GitHub repository
 - Pushing a project to GitHub
 - Pull Request workflow
-

Login/signup github account

1. Install Git

1. Download Git from the official website.
2. Website link : <https://git-scm.com/install/windows>

3. Install Git using default settings.
4. Verify the installation using the command:

```
git --version
```

2. Configure Git

Set your username and email (used for creating github account):

```
git config --global user.name "YourName"
```

```
git config --global user.email your@email.com
```

3. Clone a Repository

Follow these steps to clone an existing repository:

1. Create a new folder on your system.
2. Open the terminal inside that folder.
3. Run the clone command:

```
git clone https://github.com/naxzant/minicrm.git
```

4. Generate SSH Key

SSH keys allow secure communication between your system and GitHub.

Step 1: Generate the key

```
ssh-keygen -t ed25519 -C your@email.com
```

- Press **Enter** continuously until the process completes.

Step 2: View the public key

```
cat ~/.ssh/id_ed25519.pub
```

- Copy the displayed key and save it safely.

Note: Do not run the `ssh-keygen` command multiple times unnecessarily.

5. Push Project to GitHub

Follow these steps to push your local project:

meaning of git commands

1. `git fetch origin` → Get latest updates from remote **without changing local code**
2. `git branch` → Show all branches & current branch
3. `git add -A` → Stage **all changes** for commit
4. `git status` → Check current repo state
5. `git commit -m "msg"` → Save staged changes with a message
6. `git push -u origin branch-name` → Upload commits to remote branch

work flow:

`git fetch origin`

`git branch`

`git status`

`git add -A`

`git commit -m "Meaningful message"`

`git push -u origin branch-name`

6. Create a Pull Request (PR)

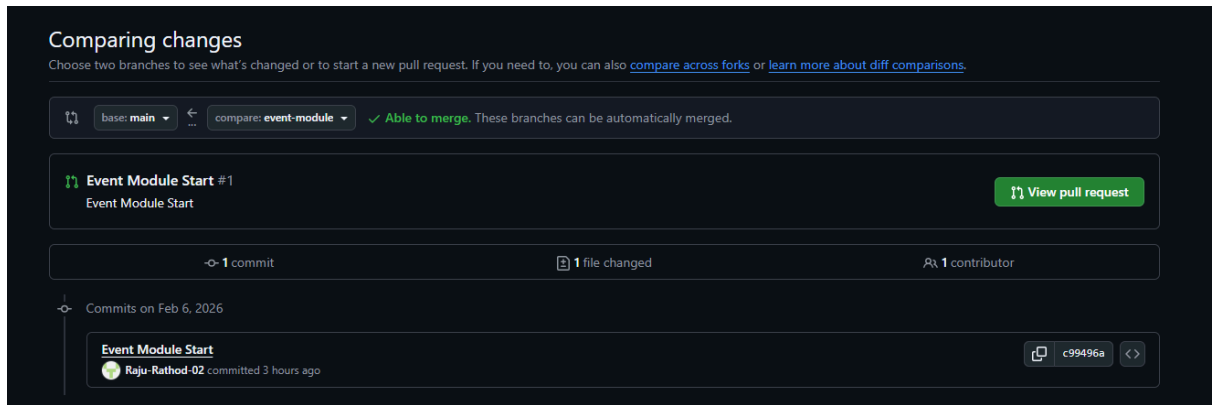
1. Create a new branch:

you will get your branch according to your crm project

2. Push the branch to GitHub:

`git push origin branch-name`

3. Open the repository on GitHub.
4. Click on pull request
5. Click **Compare & Pull Request**



- 6.
7. Add details and submit the Pull Request.

Key Concepts

- **Git:** Version control system for tracking changes
- **Repository:** A project folder tracked by Git
- **Commit:** A saved snapshot of changes
- **Branch:** A separate line of development
- **Pull Request:** A request to merge changes into another branch

Common Errors & Fixes

- **Permission denied (SSH):** Ensure the correct public SSH key is shared to GitHub
 - after sharing ssh key make sure you don't run ssh multiple times unnecessarily
 - **Git not recognized:** Reinstall Git or add Git to the system PATH(make sure you have added git to your system path)
-